

Automata and Formal Languages

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Topics

- Finite Automata
 - Deterministic Finite Automata (DFA)
 - Non-Deterministic Finite Automata (NFA)
- Extended Transition Functions
- Regular Languages
- Conversion of NFA to DFA
- Proof of Equivalence of DFA and NFA
- ϵ -Transition NFA
- ϵ -Closure
- ϵ -Transition Removal
- DFA Minimization

Finite Automata

➤ Revisiting Finite Automata for a “Switch”

➤ What are the different components?

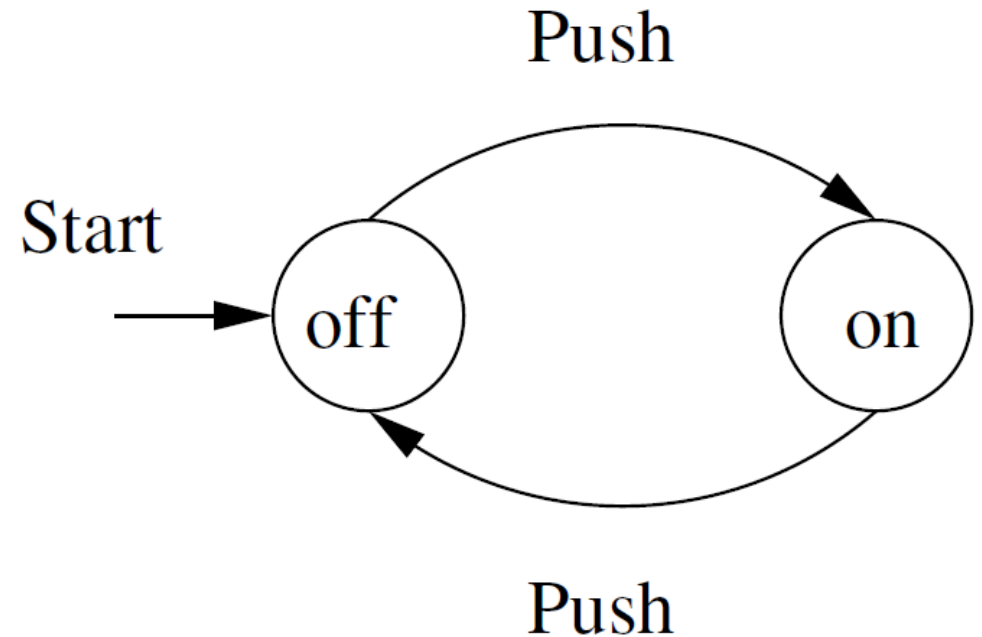
➤ States (e.g. 'off', 'on')

➤ Actions (e.g. 'Push')

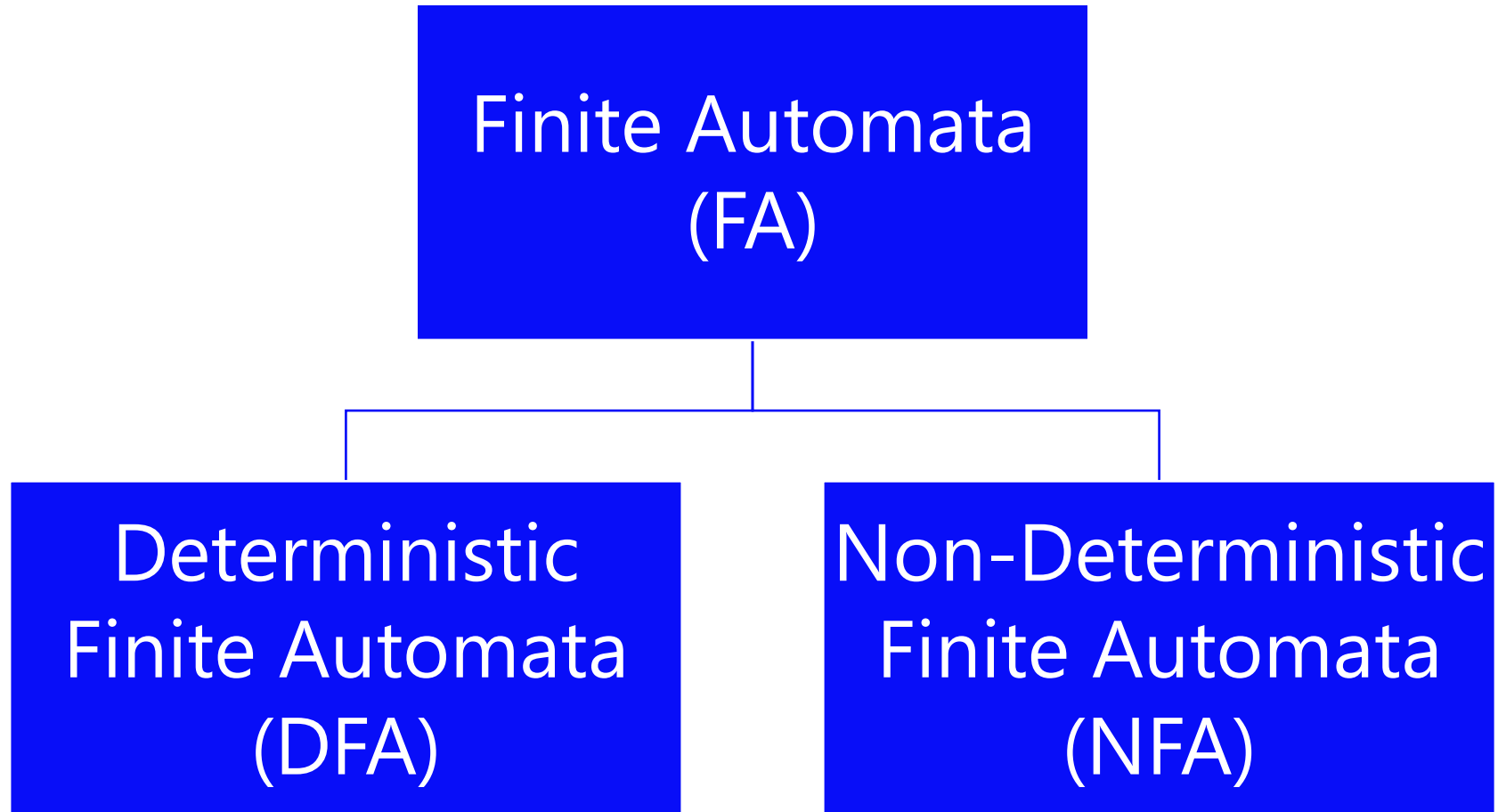
➤ Effects

➤ When in state 'off', taking action 'Push' changes the state to 'on'

➤ Start state



Finite Automata



Deterministic Finite Automata

❖ Deterministic Finite Automata consists of the following:

- A finite set of states, often denoted Q
- A finite set of input symbols, often denoted Σ
- A transition function, denoted by δ
- A start state, one of the states in Q
- A set of final or accepting states F ($F \subseteq Q$)

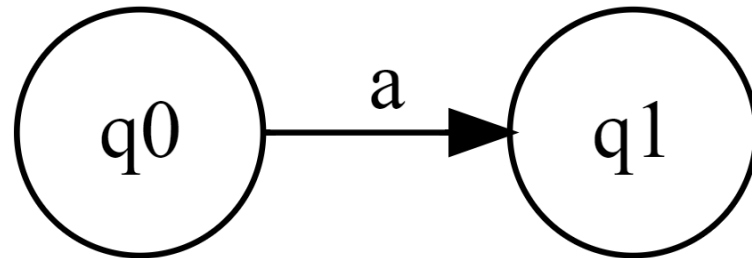
❖ Deterministic Finite Automata is represented as follows:

$$A = (Q, \Sigma, \delta, q_0, F)$$

Deterministic Finite Automata

- **Transition Function:**

$\delta(q_0, a) = q_1$ indicates that DFA enters state q_1 from state q_0 after processing input symbol a



- **Acceptance of Strings:**

For any sequence of input symbols $a_1 a_2 \dots a_i \dots a_n$ if $\delta(q_{i-1}, a_i) = q_i$ and $q_n \in F$ then the string is said to be "ACCEPTED" by the DFA else it is called "REJECTED"

Deterministic Finite Automata

□ $A = (Q, \Sigma, \delta, q_0, F)$

➤ $Q = \{q_0, q_1, q_2\}$

Can we represent it in a better way?

➤ $\Sigma = \{0,1\}$

➤ $\delta(q_0, 0) = q_1, \delta(q_0, 1) = q_0, \delta(q_1, 1) = q_1, \delta(q_1, 0) = q_2,$
 $\delta(q_2, 0) = q_2, \delta(q_2, 1) = q_2$

➤ Start state = q_0

➤ $F = \{q_2\}$

Deterministic Finite Automata

❖ DFA representations:

❖ Transition Diagram

❖ Transition Table

$$A = (\{q_0, q_1, q_2\}, \{0, 1\}, \delta, q_0, \{q_2\})$$

❖ Construction of Transition Diagram:

- For each state in Q there is a node
- For each state p in Q and each symbol a in Σ if $\delta(p, a) = q$
 - There will be an arc from p to q with label a
- There is an arrow into the start state
- Final / Accepting states are marked with double circle

Deterministic Finite Automata

□ $A = (Q, \Sigma, \delta, q_0, F)$

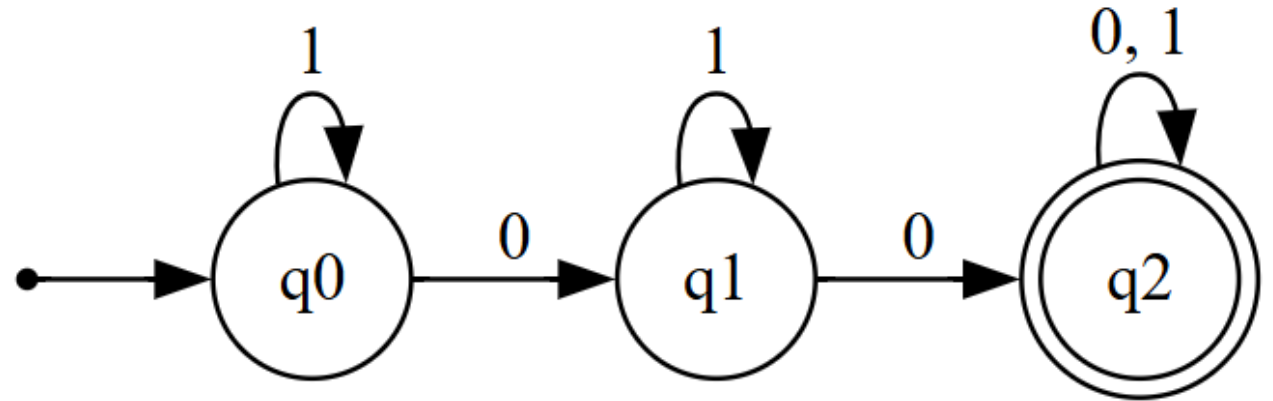
➤ $Q = \{q_0, q_1, q_2\}$

➤ $\Sigma = \{0,1\}$

➤ $\delta(q_0, 0) = q_1, \delta(q_0, 1) = q_0, \delta(q_1, 1) = q_1, \delta(q_1, 0) = q_2,$
 $\delta(q_2, 0) = q_2, \delta(q_2, 1) = q_2$

➤ Start state = q_0

➤ $F = \{q_2\}$

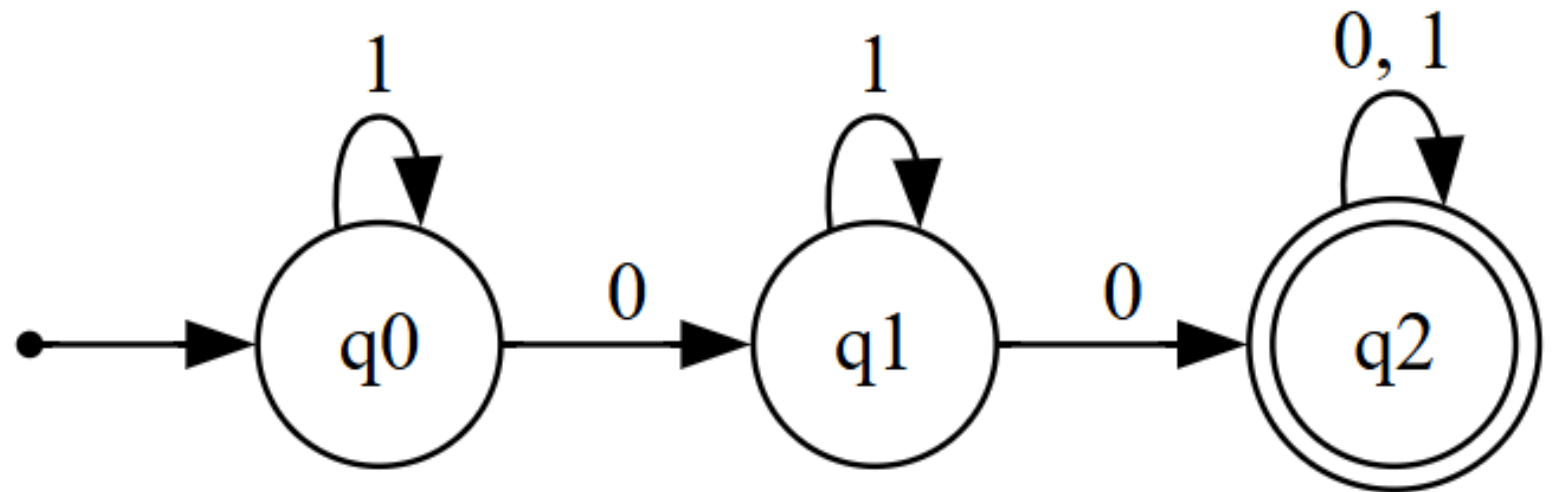


Deterministic Finite Automata

- **Transition Table:**

The entry for the row corresponding to state q and the column correspond input a is the state $\delta(q, a)$

State	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_1
$* q_2$	q_2	q_2



Deterministic Finite Automata

Extended transition Function:

$$\hat{\delta}(q, w) = p$$

- p is the state that the automaton reaches when starting in state q and processing the sequence of inputs w
- Consider $w = xa$ where a is the last symbol and x is a string then

$$\hat{\delta}(q, w) = \hat{\delta}(q, xa) = \delta(\hat{\delta}(q, x), a)$$

Extended Transition Function

$$\hat{\delta}(q_0, 010)$$

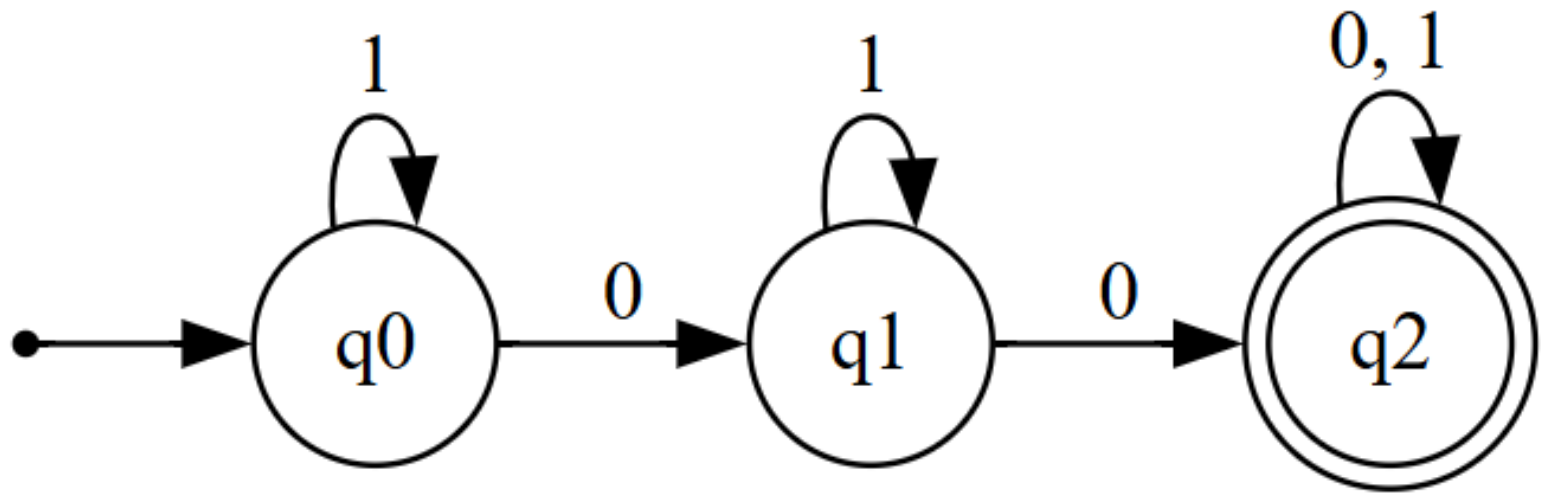
$$= \delta(\hat{\delta}(q_0, 01), 0)$$

$$= \delta(\delta(\delta(q_0, 0), 1), 0)$$

$$= \delta(\delta(q_1, 1), 0)$$

$$= \delta(q_1, 0)$$

$$= q_2$$



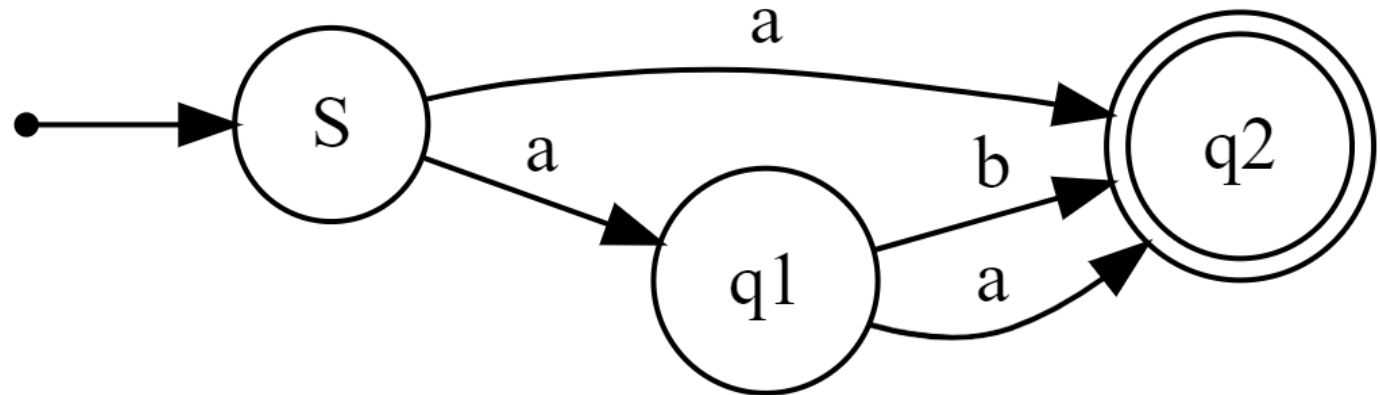
Regular Language

- Given a DFA $A = (Q, \Sigma, \delta, q_0, F)$ the **language of A** is defined as:

$$L(A) = \{w \mid \hat{\delta}(q_0, w) \text{ is in } F\}$$

- Language of the following FA?

$$L = \{a, aa, ab\}$$



Non-Deterministic Finite Automata (NFA)

❖ Non-Deterministic Finite Automata consists of the following:

- A finite set of states, often denoted Q
- A finite set of input symbols, often denoted Σ
- A transition function, denoted by δ (**Returns a subset of Q**)
- A start state, one of the states in Q
- A set of final or accepting states F ($F \subseteq Q$)

Non-Deterministic Finite Automata (NFA)

□ $A = (Q, \Sigma, \delta, q_0, F)$

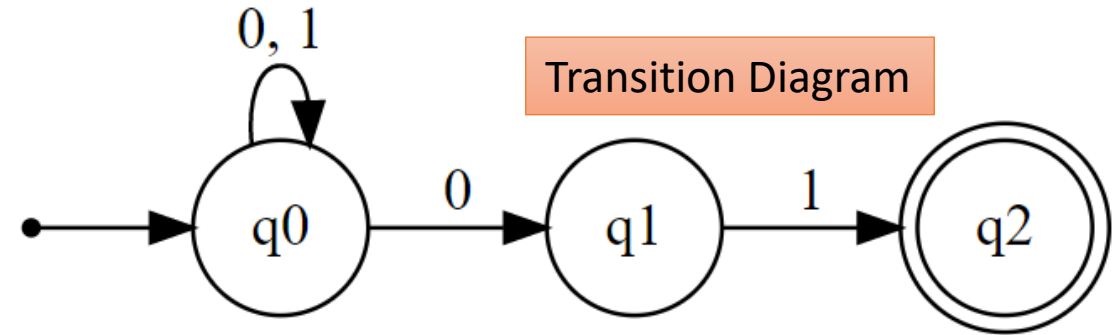
➤ $Q = \{q_0, q_1, q_2\}$

➤ $\Sigma = \{0,1\}$

➤ $\delta(q_0, 0) = \{q_0, q_1\}, \delta(q_0, 1) = q_0, \delta(q_1, 1) = q_2$

➤ Start state = q_0

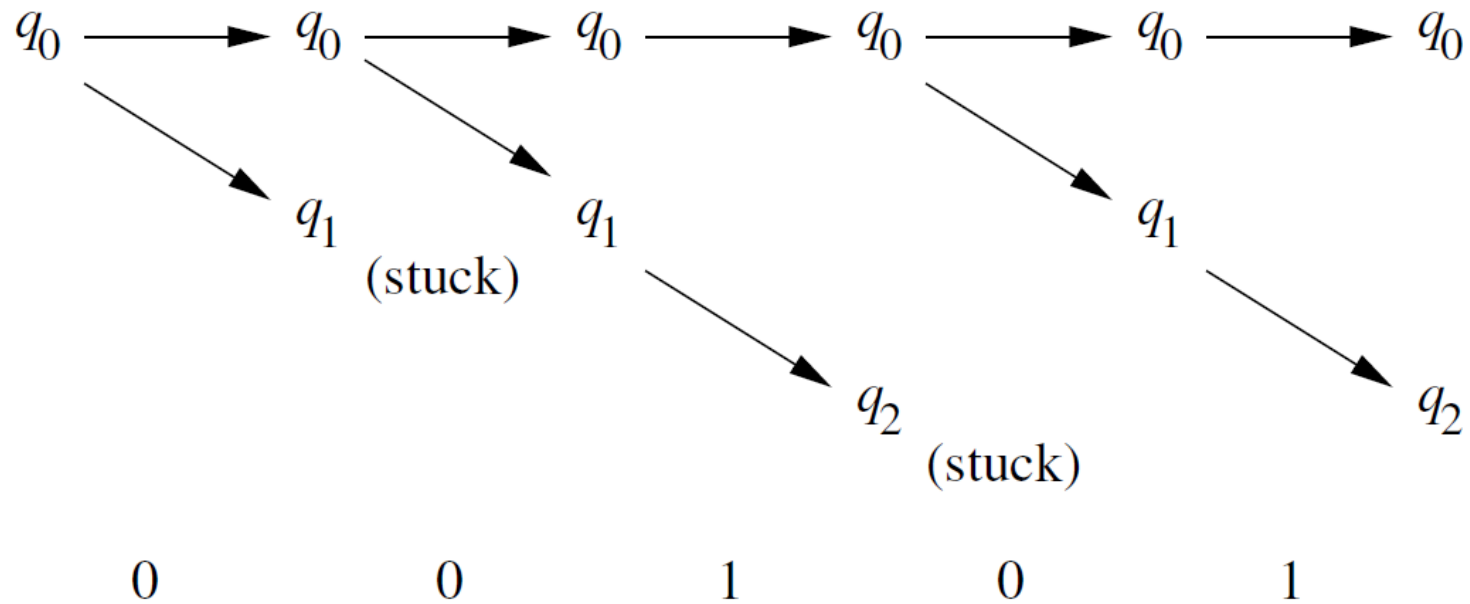
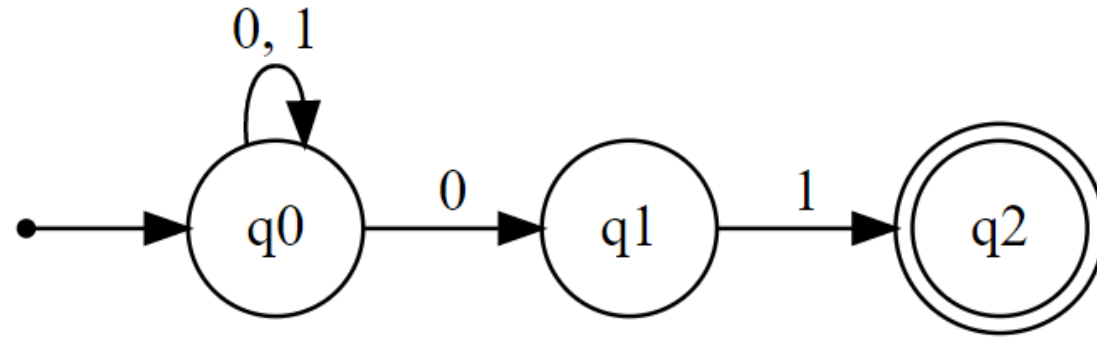
➤ $F = \{q_2\}$



Transition Table

	0	1
→ q_0	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$
$*q_2$	$\emptyset$	$\emptyset$

- How to process string 00101?



Extended Transition Function of NFA

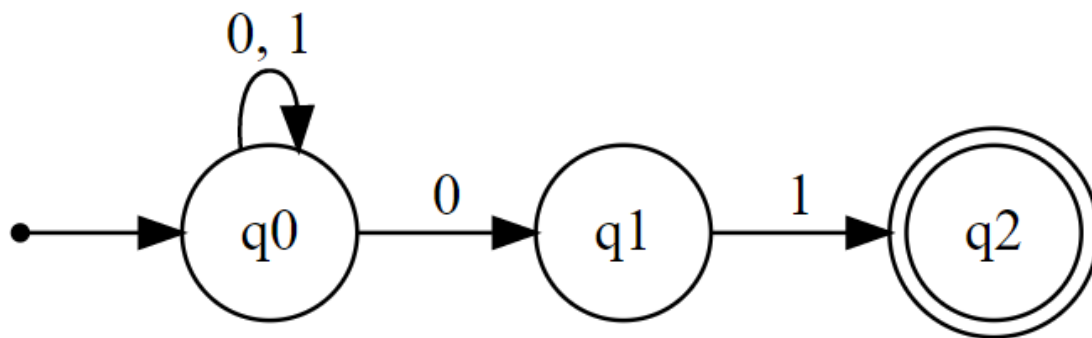
- Consider $w = xa$ where a is the last symbol and x is a rest of w .
- Assume $\hat{\delta}(q, x) = \{p_1, p_2, \dots p_k\}$

If

$$\bigcup_{i=1}^k \delta(p_i, a) = \{r_1, r_2, \dots r_m\}$$

then

$$\hat{\delta}(q, w) = \{r_1, r_2, \dots r_m\}$$



	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$
$*q_2$	$\emptyset$	$\emptyset$

$$\triangleright \hat{\delta}(q_0, 00)$$

$$= \delta(\hat{\delta}(q_0, 0), 0) = \delta(\{q_0, q_1\}, 0)$$

$$= \bigcup_{i=0}^1 \delta(q_i, 0) = \delta(q_0, 0) \cup \delta(q_1, 0)$$

$$= \{q_0, q_1\} \cup \emptyset = \{q_0, q_1\}$$

NFA to DFA Conversion

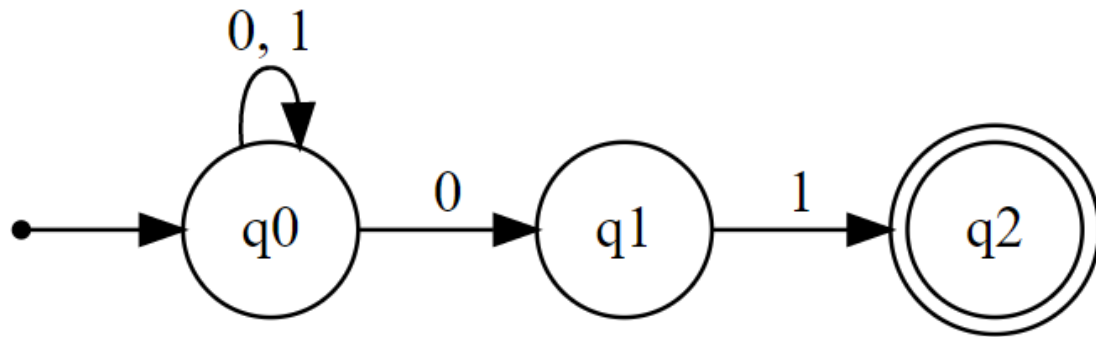
- Consider NFA $N = (Q_N, \Sigma, \delta_N, q_0, F_N)$
- Consider DFA $D = (Q_D, \Sigma, \delta_D, \{q_0\}, F_D)$
- Conversion:
 - Σ remains same
 - $Q_D \subseteq P(Q_N)$ (Note: $P(Q_N)$ is power set of Q_N)
 - For each set $S \subseteq Q_N$ and for each a in Σ

$$\delta_D(S, a) = \bigcup_{p \in S} \delta_N(p, a)$$

- Start state of D is a set that contains only the start state of N
- $F_D \subseteq P(Q_N)$ and for each $S \in F_D$, $S \cap F_N \neq \emptyset$

NFA to DFA Conversion

Subset Construction:

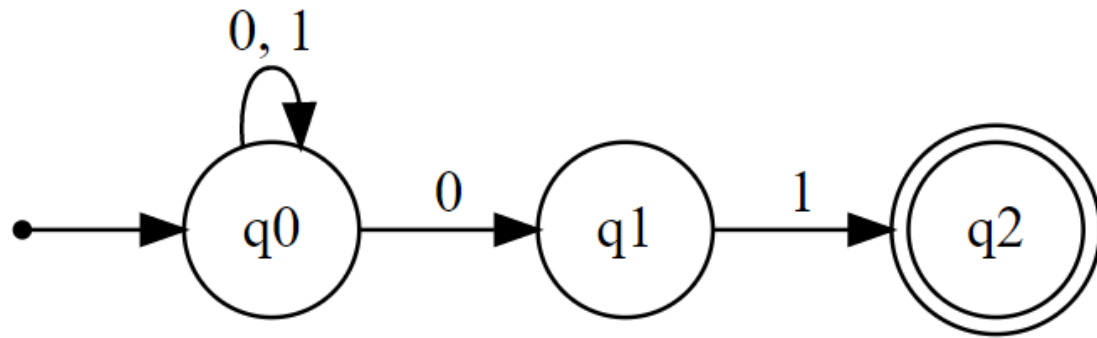


	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$
$*q_2$	$\emptyset$	$\emptyset$

	0	1
$\emptyset$	$\emptyset$	$\emptyset$
$\rightarrow \{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_1\}$	$\emptyset$	$\{q_2\}$
$*\{q_2\}$	$\emptyset$	$\emptyset$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$*\{q_0, q_2\}$	$\{q_0, q_1\}$	$\{q_0\}$
$*\{q_1, q_2\}$	$\emptyset$	$\{q_2\}$
$*\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$

Can we improve NFA to DFA conversion?

NFA to DFA Conversion

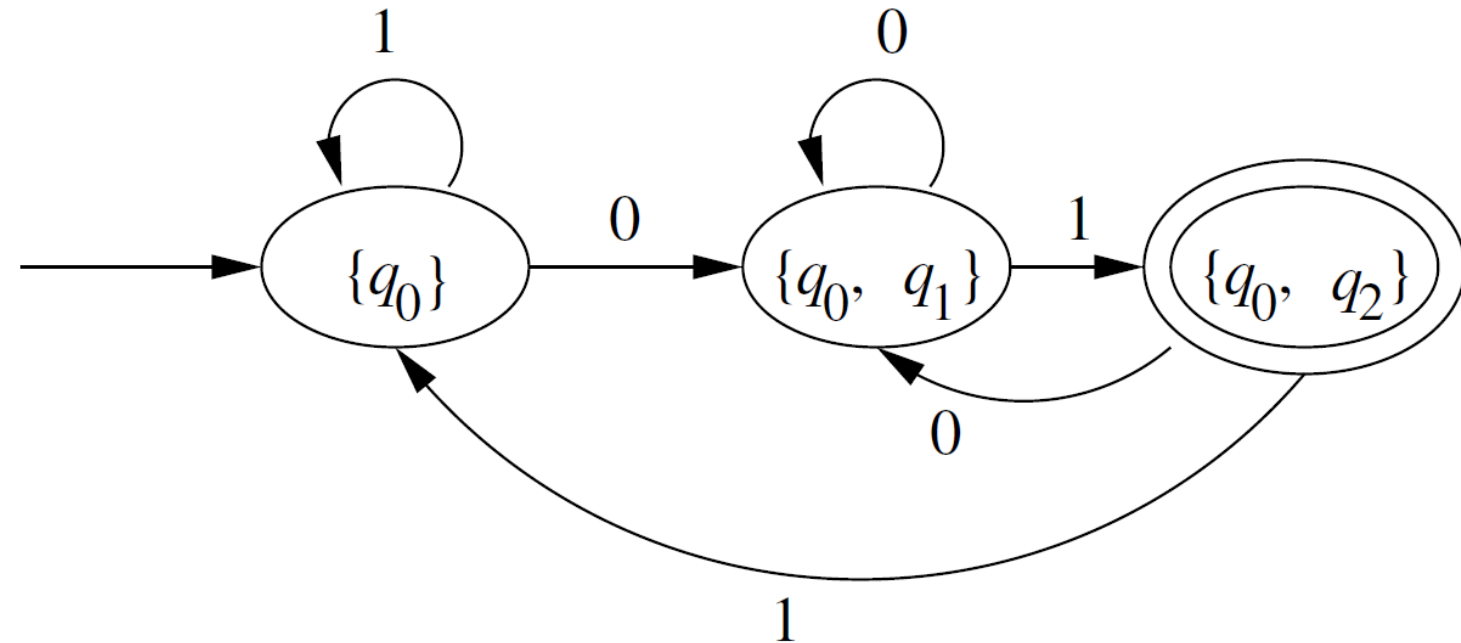


	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\emptyset$	$\{q_2\}$
$*q_2$	$\emptyset$	$\emptyset$

	0	1
$\rightarrow \{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$*\{q_0, q_2\}$	$\{q_0, q_1\}$	$\{q_0\}$

NFA to DFA Conversion

	0	1
$\rightarrow \{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$*\{q_0, q_2\}$	$\{q_0, q_1\}$	$\{q_0\}$



Equivalence of NFA and DFA

If DFA $D = (Q_D, \Sigma, \delta_D, \{q_0\}, F_D)$ is constructed from NFA $N = (Q_N, \Sigma, \delta_N, q_0, F_N)$ using subset construction then $L(D) = L(N)$

Proof Outline:

- Show the following by induction on $|w|$

$$\hat{\delta}_D(\{q_0\}, w) = \hat{\delta}_N(q_0, w)$$

Equivalence of NFA and DFA

- **Basis**

Consider $|w| = 0$, then $w = \epsilon$

Both $\hat{\delta}_D(\{q_0\}, \epsilon)$ and $\hat{\delta}_N(q_0, \epsilon)$ are $\{q_0\}$

Hence, $\hat{\delta}_D(\{q_0\}, w) = \hat{\delta}_N(q_0, w)$

- **Induction**

Assume that the $\hat{\delta}_D(\{q_0\}, w) = \hat{\delta}_N(q_0, w)$ holds for $|w| = n$

Consider, $|w| = n + 1$ and $w = xa$ where a is the final symbol of w

Hence, $|x| = n$ thus by induction hypothesis $\hat{\delta}_D(\{q_0\}, x) = \hat{\delta}_N(q_0, x)$

Equivalence of NFA and DFA

Consider that $\hat{\delta}_D(\{q_0\}, x) = \hat{\delta}_N(q_0, x) = \{p_1, p_2, \dots, p_k\}$

Further we get;

$$\hat{\delta}_N(q_0, w) = \hat{\delta}_N(q_0, xa) = \delta_N(\hat{\delta}_N(q_0, x), a) = \bigcup_{i=1}^k \delta_N(p_i, a)$$

Now, from subset construction we get;

$$\delta_D(\{p_1, p_2, \dots, p_k\}, a) = \bigcup_{i=1}^k \delta_N(p_i, a)$$

Equivalence of NFA and DFA

$$\hat{\delta}_D(q_0, w) = \hat{\delta}_D(\{q_0\}, xa) = \delta_D(\hat{\delta}_D(\{q_0\}, x), a)$$

$$= \delta_D(\{p_1, p_2, \dots, p_k\}, a) = \bigcup_{i=1}^k \delta_N(p_i, a)$$

Both D and N accepts w iff $\hat{\delta}_D(\{q_0\}, w)$ and $\hat{\delta}_N(q_0, w)$ contains a state in F_N .

Hence, $L(D) = L(N)$